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A Supplementary Methods

A.1 Agentic protein-design harness

OpenDDE-Harness formulates antibody design as a constrained, history-dependent search over CDR sequence space. The system separates three roles: LLM-based parent selection and CDR proposal, OpenDDE-based molecular evaluation, and deterministic population control. Agents propose and interpret designs; the harness enforces framework constraints, executes evaluation, updates the population, and records the complete design trajectory.

A.1.1 Problem setting

Given an antigen structure, an epitope specification g , a fixed antibody framework, and designable CDR positions $\mathcal{C}$, the objective is to generate antibody sequences with favorable predicted antigen engagement while preserving framework integrity and developability. A variable domain follows the structure

$$\text{FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4},$$

where $\mathcal{C}$ contains the editable positions. Fab designs may include heavy- and light-chain variable domains, whereas nanobody designs use a single heavy-chain variable domain.

A.1.2 OpenDDE-Harness framework

At cycle t , the search state is

$$s_t = (\mathcal{P}_t, \mathcal{H}_t, r_t, g),$$

where $\mathcal{P}_t \subset \mathcal{X}$ is the current antibody population, $\mathcal{H}_t$ is the design history, r_t is the current reflection context, and $\mathcal{X}$ is the set of candidates compatible with the fixed framework and editable CDR positions.

The history records parent candidates, CDR edits, generated variants, predicted complexes, molecular observations, gate outcomes, and selection status. It includes both successful and failed candidates.

The parent-selection agent selects a parent from the current population:

$$x_t \sim \pi_\alpha(x \mid \mathcal{P}_t, \mathcal{H}_t, r_t, g),$$

where π_α uses candidate quality, lineage, diversity, and prior design evidence to balance exploitation and exploration.

Conditioned on the selected parent and the current design context, the CDR design agent proposes B edit actions:

$$a_t^{(1)}, \dots, a_t^{(B)} \sim \pi_\theta(a \mid x_t, g, \mathcal{C}, \mathcal{H}_t, r_t).$$

An action may specify a point substitution, insertion, deletion, localized mutation set, or full CDR redesign. Applying an action produces

$$\tilde{x}_{t+1}^{(b)} = a_t^{(b)}(x_t), \quad b = 1, \dots, B.$$

The harness validates each proposal against amino-acid, framework, CDR, and mutation-budget constraints before evaluation.

Each valid candidate is evaluated by OpenDDE and auxiliary computational tools:

$$\left(\hat{y}_{t+1}^{(b)}, o_{t+1}^{(b)}\right) \sim \hat{P}_\phi\left(\cdot \mid \tilde{x}_{t+1}^{(b)}, g\right),$$

where $\hat{y}_{t+1}^{(b)}$ is the predicted antibody–antigen complex and $o_{t+1}^{(b)}$ is the resulting observation vector. The observations may include structural confidence, interface quality, epitope engagement, interaction features, and developability properties. These values are computational proxies and are not direct measurements of affinity.

The new records are appended to the history:

$$\mathcal{H}_{t+1} = \mathcal{H}_t \cup \left\{ \left(x_t, a_t^{(b)}, \tilde{x}_{t+1}^{(b)}, \hat{y}_{t+1}^{(b)}, o_{t+1}^{(b)} \right) \right\}_{b=1}^B.$$

The population controller then applies validation, gating, ranking, duplicate removal, elitist retention, and diversity criteria:

$$\mathcal{P}_{t+1} = \Gamma_K \left(\mathcal{P}_t \cup \{\tilde{x}_{t+1}^{(b)}\}_{b=1}^B, \mathcal{H}_{t+1} \right),$$

where Γ_K retains at most K candidates. Filtering is therefore part of population control, not a standalone reward function.

At the configured reflection interval, the reflection agent summarizes the updated history and population:

$$r_{t+1} \sim \pi_\beta (r \mid \mathcal{H}_{t+1}, \mathcal{P}_{t+1}, g).$$

The resulting context identifies beneficial or harmful mutations, interaction patterns, structural failure modes, developability liabilities, lineage trends, and proposed search directions. It may also specify target CDR positions, mutation budgets, preserved constraints, and exclusions for the next cycle:

$$x_{t+1} \sim \pi_\alpha (x \mid \mathcal{P}_{t+1}, \mathcal{H}_{t+1}, r_{t+1}, g),$$

$$a_{t+1} \sim \pi_\theta (a \mid x_{t+1}, g, \mathcal{C}, \mathcal{H}_{t+1}, r_{t+1}).$$

The reflection context is a derived cycle-level control signal, whereas $\mathcal{H}_t$ remains the persistent record of the search. Thus, OpenDDE-Harness does not optimize a fixed weighted reward. It combines LLM-based proposal and interpretation with explicit constraints, OpenDDE-based evaluation, deterministic population updates, and persistent provenance to prioritize antibody candidates for high-affinity binder validation.

A.2 Population database

The population database is a bounded archive that supports parent selection, lineage tracking, and diversity preservation. By maintaining multiple antibody lineages, OpenDDE-Harness can exploit high-quality candidates while avoiding premature convergence to a single sequence family.

Before entering the population, candidates must be successfully folded and scored, contain no unresolved residues, and satisfy the structure-based admission criteria in Table 1. These gates promote an antibody-like binding pose by directing CDR contacts toward the target and, when specified, CDR3 contacts toward the intended epitope, while minimizing undesired contacts between the antibody framework and the target. Candidates that fail the gates are excluded from the formal population, although a promising candidate may temporarily serve as a working parent to maintain search continuity.

The population follows a constrained-elite strategy. Existing and newly admitted candidates are jointly ranked by the optimization objective, deduplicated, and retained subject to CDR sequence-diversity and mutation-reuse constraints. Parents are subsequently selected using fitness-weighted sampling with a small exploratory component, balancing the exploitation of strong candidates with continued exploration of alternative antibody lineages.

Table 1: *Epitope specific gate for population updates.*

Criterion	Threshold	Description
CDR3–epitope contact	≥ 1	Requires at least one CDR3–epitope heavy-atom contact within 5.0 Å, enforcing direct engagement of the intended antigen region.
CDR interface fraction	≥ 0.50	Requires CDR residues to contribute at least half of the predicted interface contacts, limiting framework-dominated and nonspecific poses.

A.3 Sequence optimization

LLM-proposed sequences may depart from the distribution of naturally occurring antibody sequences. Following Germinal [6] and BindCraft [8], candidates that passed the initial structural gates were optimized with SolubleMPNN [37] to improve sequence plausibility, solubility, and expression-related properties.

SolubleMPNN generated 40 sequences for each design. The four highest-ranked sequences were retained for OpenDDE refolding and final filtering. To preserve the predicted binding mode, optimization was restricted to CDR residues whose heavy atoms were more than 3 Å from the antigen. Residues at the predicted interface were therefore held fixed during sequence optimization.

A.4 Filtering for experimental validation

To prioritize optimized antibody candidates for experimental validation, we applied an agent-guided ranking stage after refolding with OpenDDE. Each candidate was evaluated using a comprehensive panel of metrics, including ipTM, pLDDT, ranking score and binder RMSD. These metrics provided complementary evidence concerning prediction confidence, structural consistency, and the predicted binding geometry.

The candidate-level measurements, together with the design objectives and metric definitions, were provided to the filter-agent for comparative assessment. Rather than applying a fixed weighted score, the agent ranked candidates by considering the agreement and trade-offs among the available metrics, while accounting for overlapping confidence signals and explicitly noting missing evidence. The agent returned an ordered shortlist with candidate-specific selection rationales and potential risks to guide subsequent experimental validation.

A.5 LLM selection

We compared LLMs for agentic sequence-design reasoning on IL3 (Table 2). Claude Sonnet 4.6 achieved the highest success rate (18%), whereas GPT-5.5 produced the highest single-candidate ipTM (0.923). Both required substantially more inference time and estimated token cost than Gemini-3 Flash. Gemini-3 Flash achieved an 8% success rate and a best ipTM of 0.898 with a logged inference time of 129.1 and an estimated cost of \$0.046 per run. It was approximately 1.8× faster and 3.4× less expensive than Claude Sonnet 4.6, and approximately 7× less expensive than GPT-5.5. Relative to Gemini-3.1 Flash Lite, Gemini-3 Flash doubled the success rate and increased the best ipTM from 0.870 to 0.898.

We selected Gemini-3 Flash as the default model because it provided the best trade-off between computational success, structural quality, runtime, and cost for repeated design and reflection cycles.

Table 2: Language-model comparison for IL3 design reasoning. All runs used matched prompts, a 100-cycle budget, identical hotspot constraints, and OpenDDE evaluation. Success rate denotes the fraction of 100 retained candidates with $iPTM \geq 0.80$. Estimated costs use logged input, output, and cache-read tokens priced at public list rates.

Model	Success rate	Best iPTM	Inference time	Estimated cost (USD)
Gemini-3 Flash	8%	0.898	129.1	0.046
Gemini-3.1 Flash Lite	4%	0.870	60.9	0.006
GPT-5.5	2%	0.923	219.7	0.322
Claude Sonnet 4.6	18%	0.909	235.4	0.156

A.6 Prompts for design agent

For sequence design, we used an LLM-based design agent to propose CDR-localized edits through multi-aspect molecular reasoning. The agent conditions on the selected parent antibody, target context, designable CDR positions, previous design history, reflection context, and structural and developability constraints. Its output consists of either concrete mutation specifications or redesigned CDR sequences, depending on the design mode. The system prompt encodes antibody-specific priors, including framework architecture, CDR roles, antigen-recognition principles, constructability constraints, and mutation-risk control. Figure 7 shows the system prompt used for the design agent.

Design Instructions
You are the Design Agent for an antibody-design pipeline. You plan feasible candidates, then emit only executable designs. INVIOABLE CONSTRAINTS - Mutable positions: mutate only positions explicitly marked mutable. - Fixed residues: never alter residues marked fixed. - Candidate count: emit exactly the required number of candidates. - Mutation count: each candidate must stay within the allowed mutation budget. - Output schema: emit only schema-valid output. No extra keys, commentary, or markdown. Never relax, reinterpret, approximate, round, or partially waive these constraints. Scientific rationale, user preference, prior context, or “conservative substitutions” do not override them.

Figure 7: System prompt for design agent.

A.7 Prompts for reflection agent

The reflection agent interprets molecular observations from the world-model evaluation together with the accumulated design history. Its role is not to generate antibody sequences directly, but to produce a structured reflection context that summarizes successful design patterns, recurrent failure modes, developability concerns, and candidate search directions for the next design cycle. This reflection context is then provided to the parent-selection and design agents to guide subsequent parent choice and CDR editing. Figure 8 displays the system prompt used for the reflection agent.

Reflection Instructions

You are an Antibody Design Critic. You combine short-horizon reflection on the latest folding cycle with long-horizon reasoning over the full evolutionary history of antibody variants. Your job is to produce a grounded markdown reflection report for the current cycle, attributing successes and failures to specific CDRs and antibody-engineering mechanism classes.

Core rules:

- Use only evidence from the current prompt and any tool or script output produced in this run.
- Prefer exact candidate names and exact metric values from the provided results.
- Do not invent candidate names, metrics, mutations, structural contacts, or evolutionary findings.
- Treat chat history and memories as non-authoritative hints, not evidence.
- If current-run structure analysis is unavailable, explicitly say structural conclusions are unavailable.
- Read the Antibody Identification Header (Section 0 of the analyze-phase report) for chain roles (VH/VL/VHH) and CDR ranges; use those names everywhere you attribute mutations.
- The hallucinate phase `strategy` field uses a `[CDR / mechanism\_class] ...` prefix convention; you may parse this prefix to attribute candidates to a CDR and mechanism. If the prefix is missing or malformed, attribute by looking up the mutation positions in the Header's CDR ranges.
- Follow the report format requested in the user prompt. Do not output JSON unless the user prompt explicitly asks for JSON.

Figure 8: System prompt for reflection agent.

A.8 Biolayer interferometry

Binding of antibody candidates to TfR1 was assessed by biolayer interferometry (BLI) using Protein A biosensors. Recombinant TfR1 (catalog no. 11020-H01H) was loaded onto the biosensors at 20 $\mu\text{g}/\text{mL}$ for 500 s. Assays were performed in $2\times$ PBST supplemented with 0.1% BSA and 0.02% Tween 20. Antibody candidates were tested at five concentrations in a twofold dilution series from 1000 to 62.5 nM, with association and dissociation monitored for 120 and 180 s, respectively.

A.9 Surface plasmon resonance

Antibody-antigen binding kinetics were measured using a Cathera LSA XT instrument equipped with a PAGHC30M sensor chip. Antibodies were diluted to 2 $\mu\text{g}/\text{mL}$ in $1\times$ HBS-EP+ running buffer and captured on the sensor surface for 10 min. Antigens were prepared as threefold serial dilutions in running buffer, with a maximum concentration of 300 nM, and injected in ascending concentration order. Buffer-only injections preceded the antigen series to stabilize the baseline. Each antigen injection comprised a 60 s baseline, a 300 s association phase, and a 900 s dissociation phase. Following each antigen titration series, the sensor surface was regenerated with three 20 s pulses of 10 mM glycine-HCl, pH 2.0. Sensorgrams were analyzed using Cathera KIT software and fitted to a 1:1 Langmuir binding model to estimate the association and dissociation rate constants (k_{on} and k_{off}). The equilibrium dissociation constant was calculated as $K_D = k_{\text{off}}/k_{\text{on}}$.

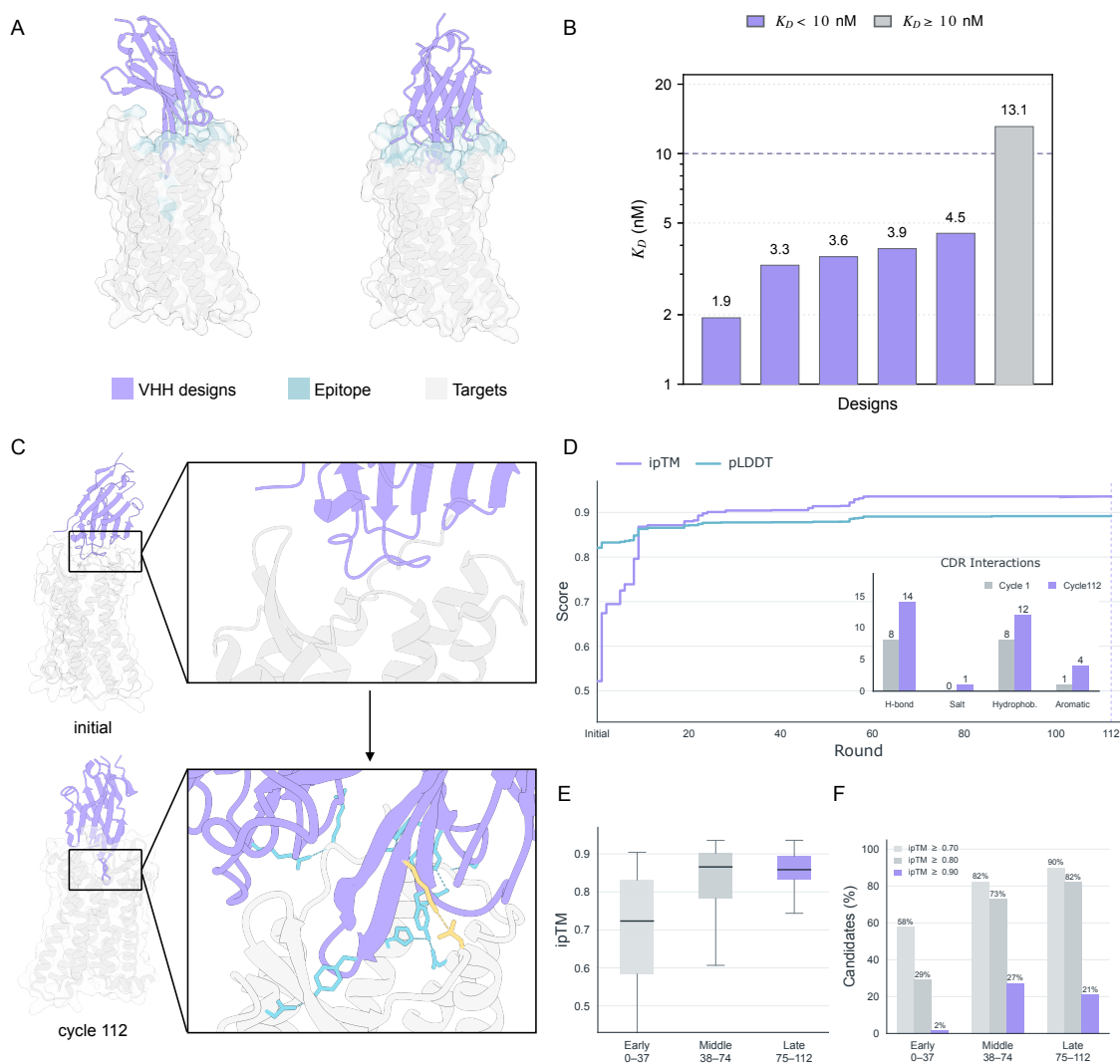


Figure 9: Computational progression of a CXCR4 VHH lineage. (A) Two predicted CXCR4-binding modes. VHH designs are purple, epitope residues are cyan, and CXCR4 is grey. (B) Measured K_D values for six binders. Five exhibit $K_D < 10$ nM (purple); one exhibits $K_D \geq 10$ nM (grey). The dashed line marks 10 nM. (C) Initial and cycle-112 predicted complexes with enlarged interface views. Annotated hydrogen bonds and salt bridges are cyan and orange, respectively. (D) Running maxima of ipTM and pLDDT over 112 design cycles. The inset compares predicted CDR interaction counts at cycles 1 and 112. (E) ipTM distributions during early (rounds 0–37), middle (38–74), and late (75–112) search stages. (F) Candidate fractions meeting ipTM thresholds of ≥ 0.70 , ≥ 0.80 , and ≥ 0.90 within each stage. Running maxima cannot decrease; stage-wise distributions summarize the broader candidate population. Complex structures and contacts are predictions; experimental binding was measured for a SolubleMPNN-refined descendant of the illustrated lineage.

Table 3: Target information for the *in silico* benchmark.

Target	Antigen description	Format	Length	Hotspots
AHSP	Alpha-hemoglobin stabilizing protein	scFv	102	None
BHRF1	Epstein–Barr virus Bcl-2 homolog BHRF1	VHH	149	A72, A93, A99, A100, A102
IL-20	Interleukin-20	VHH	152	A7, A8, A9
IL-3	Interleukin-3	VHH	112	A23, A25, A26, A31, A40, A104
PD-L1	Programmed death-ligand 1	VHH	115	A37, A39, A41, A96, A98
RNF43	Ring finger protein 43 ectodomain	scFv	155	None

B Supplementary Results

B.1 In-silico Benchmark Setting

We assembled an *in silico* benchmark of six antigen targets spanning distinct binding contexts (Table 3). Target-specific hotspots were supplied as design constraints when available; targets without hotspots were unconstrained. We compared OpenDDE–Harness with Germinal and HalluDesign, together with a direct LLM sequence-generation baseline. Each method generated 200 designs per target from the same antigen context. All pipeline failures were retained in the accompanying source data.

For OpenDDE–Harness, Germinal, and HalluDesign, each designed complex was optimized with SolubleMPNN. Framework residues and CDR residues with any heavy atom within 3 Å of the antigen were fixed; only non-interface CDR residues were redesigned. SolubleMPNN sampled 40 sequences per design at a temperature of 0.2. Sequences that changed fixed residues were removed, duplicates were collapsed, and the four highest-scoring sequences were retained. The direct LLM baseline bypassed SolubleMPNN and was refolded from its original generated sequences.

All retained sequences were refolded once with OpenDDE using antigen MSAs and no template information. For each design, the parsed refold with the highest ipTM was selected. Structural self-consistency was measured as binder C_{α} RMSD between the selected refold and the designed structure after Kabsch superposition of matched antibody C_{α} atoms. A design was considered successful when its selected refold satisfied ipTM > 0.75, pLDDT > 0.85, and binder C_{α} RMSD < 2.0 Å.

We quantified CDR diversity by clustering concatenated CDR sequences with MMseqs2 [41] easy-cluster at 80% sequence identity and 80% coverage. For Fig. 5B, clustering used successful sequences only. For Fig. 5C, clustering used all generated sequences; a cluster was successful if it contained at least one successful design. The cluster-level success rate is the fraction of successful clusters among all clusters.

Sequence novelty was assessed for successful designs only. We calculated the minimum Levenshtein distance between each concatenated CDR sequence and its nearest SAbDab sequence after removing CDR delimiters (Fig. 10). Larger distances indicate greater sequence novelty relative to the SAbDab reference repertoire.

For Fig. 5D, we report ipTM, pLDDT, and ranking-score distributions for the top-ipTM parsed refold of every design, irrespective of its final success label. Dashed lines denote ipTM = 0.80, pLDDT = 0.85, and ranking score = 0.80. Figure 11 reports binder C_{α} RMSD for the same top-ipTM refolds, with the 2.0 Å threshold shown for reference.

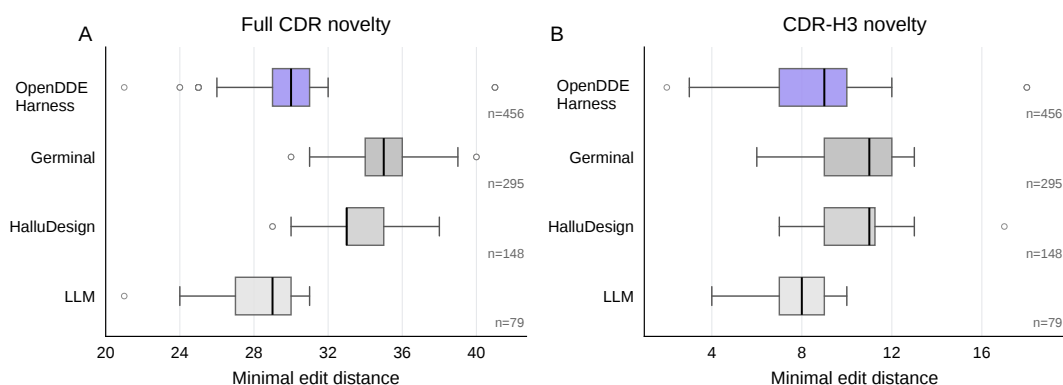


Figure 10: CDR sequence novelty among successful designs. Novelty is the minimum Levenshtein distance between each successful design and its nearest SABDab sequence. (A) Concatenated CDR sequences. (B) CDR-H3 sequences. Boxes show the interquartile range, centre lines show medians, whiskers extend to 1.5× the interquartile range, and points denote individual designs. n is the number of successful designs.

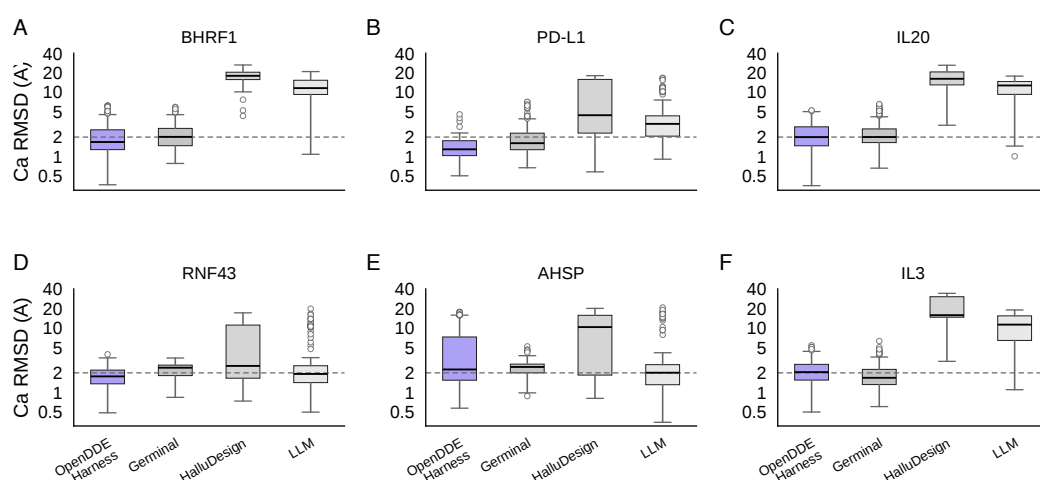


Figure 11: Folding consistency across six targets. (A-F) Binder C_{α} RMSD after antibody-based superposition on six targets. Boxes show quartiles and whiskers extend to 1.5 times the interquartile range. Panels are retained from the source; the success-selection rule, denominators, and score-panel sampling require the audit in Author check 3. The direct LLM baseline received fewer sequence-optimization and refolding opportunities.

B.2 In-silico Evaluation of Antibody Developability

High predicted binding scores do not guarantee antibody developability. We therefore evaluated re-fold-successful designs using six computational metrics: CDR patch surface hydrophobicity (PSH), positive and negative patch charge (PPC and PNC), spatial charge symmetry (SFvCSP), isoelectric point (pI), and the number of predicted CDR sequence-liability sites (Fig. 12).

Compared with Germinal and HalluDesign, OpenDDE-Harness produced lower PSH values, indicating reduced hydrophobic patch formation (Fig. 12A). Its PPC, PNC, and SFvCSP distributions were also concentrated near zero with fewer extreme values, consistent with more balanced CDR electrostatics (Fig. 12B–D).

The generated antibodies showed a narrow pI distribution of approximately 8.0–8.8 and a lower CDR liability burden than HalluDesign (Fig. 12E–F). These results indicate that the optimization procedure improved predicted interface properties without substantially increasing the screened developability

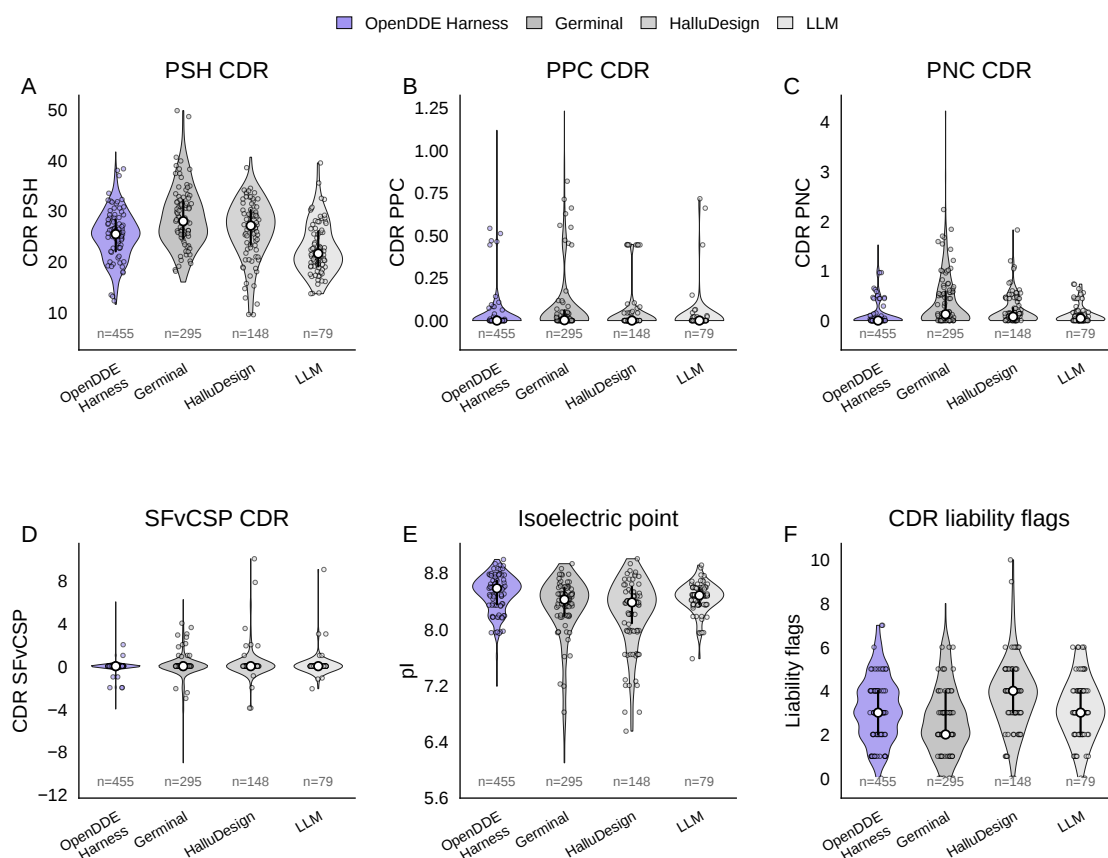


Figure 12: In silico developability of refold-successful antibody designs. Violin plots show six developability metrics for each method. (A) CDR patch surface hydrophobicity (PSH); lower values indicate reduced hydrophobic patch formation. (B) Positive patch charge (PPC). (C) Negative patch charge (PNC). (D) Spatial charge symmetry (SFvCSP); lower values indicate more balanced charge distribution. (E) Isoelectric point (pI); extreme values can increase developability risk. (F) The number of predicted sequence-liability sites within the CDRs; lower values indicate fewer potential sequence liabilities. Sample sizes n denote the number of refold-successful designs per method and therefore differ across methods. Boxes indicate the interquartile range, central markers indicate the median, and faint points show individual designs.

risks. The reported metrics are computational proxies; they do not establish expression, aggregation behavior, pharmacokinetics, or experimental therapeutic suitability.

B.3 Throughput analysis

Using observed per-trajectory completion times on a single 80 GB A100 GPU, we estimated the number of unique antibody designs produced in 24 h before final experimental filtering. Hallucination-based methods return one candidate per optimization trajectory. OpenDDE-Harness updates a population of candidates within each trajectory and returns a candidate library.

Across the six targets, OpenDDE-Harness produced a mean of 537 designs per GPU-day, compared with 228 for HalluDesign and 207 for Germinal (Table 4). This corresponds to approximately $2.4\times$ the throughput of HalluDesign and $2.6\times$ that of Germinal.

Table 4: Estimated design throughput on a single 80 GB A100 GPU. Throughput is the estimated number of unique antibody designs generated in 24 h before final experimental filtering, extrapolated from observed per-trajectory completion times.

Target	OpenDDE-Harness	Germinal	HalluDesign
AHSP	344	187	216
BHRF1	570	228	136
IL20	701	220	142
IL3	410	323	334
PD-L1	782	244	331
RNF43	417	40	210
Mean	537	207	228